

USER GUIDE

Community-Based Sarcopenia Screening Tool for Older Adults

Illustrated instructions for batch screening, result interpretation, and safe use

Research and preliminary screening use only. This tool does not diagnose sarcopenia and must not be used as the sole basis for treatment or referral decisions.

Document item	Current release
Document version	1.0
Web application version	1.1.0
Locked model version	1.0.0
Last updated	August 2026
Screening mode	Batch upload using CSV or XLSX

Intended readers

Trained community-health, primary-care, and research personnel who conduct preliminary sarcopenia risk screening among adults aged 65 years and older.

Public resources

Source code, synthetic examples, automated tests, validation evidence, and documentation are available at <https://github.com/wenzisgjm/community-sarcopenia-screening-tool>.

1. Purpose, scope, and safety

The tool supports preliminary sarcopenia risk screening in community or primary-care settings. It identifies participants who should be considered for further standardised assessment; it does not establish a diagnosis.

Users can upload batch data in CSV or Excel format. The current release does not provide manual single-participant data entry.

Community-Based Sarcopenia Screening Tool for Older Adults

Strong bodies build strong communities.

Target users: adults aged 65 years and older.

Use case: preliminary sarcopenia risk screening in community or primary care settings.

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Batch Screening

Users can upload batch data in CSV or Excel format. The file must contain participant codes, sex, and the required screening variables. The tool will identify screen-positive participants who should receive further assessment.

Upload a CSV or Excel file containing at least the following columns:

ID, sex, age, EQ5D, pa_aerobic, is_allownc, Chew_Diff, Prot_Deficiency, N_EN, HE_wc, obe_4class

Note: ID and sex are retained for participant identification and result review. They are not used to calculate model risk.

Select a file for batch prediction

[Upload](#) 200MB per file • CSV, XLSX

Figure 1. Batch-screening landing page, intended-use statement, safety warning, and required columns.

Before use

- Understand the variable definitions and coding rules in this guide.
- Understand the difference between screening and diagnosis.
- Know the local procedures for confirmatory assessment and referral.
- Remove unnecessary direct identifiers before uploading data.
- Confirm that institutional policy permits use of the selected hosting environment.

2. Quick-start workflow

- 1 Download the CSV template from the How to Use tab or use examples/sample_input.csv.
- 2 Enter one participant per row and retain the required column names exactly.
- 3 Confirm that all participants are aged 65 years or older.
- 4 Confirm that categorical variables use the required numeric codes.
- 5 Open the Screening tab and upload the CSV or XLSX file.
- 6 Review warnings or errors. Correct the source file and upload it again when an error is displayed.
- 7 Review the screening summary and participant-level results.
- 8 Download the result CSV and store it according to institutional policy.
- 9 Arrange standardised assessment for screen-positive participants according to the local protocol.

Supported files

- CSV (.csv)
- Excel Workbook (.xlsx)
- Legacy Excel .xls files are not supported; save them as .xlsx or .csv before upload.

Required column order

ID, sex, age, EQ5D, pa_aerobic, is_allownc, Chew_Diff, Prot_Deficiency, N_EN, HE_wc, obe_4class

Data preparation principles

- Use a non-identifying participant code in ID.
- Keep column names unchanged and use one participant per row.
- Use numeric values and the specified numeric category codes.
- Do not enter free-text category labels in model-input columns.
- Check implausible values and missingness before upload.

3. Required input data

Column	Definition and permitted coding
ID	Non-identifying participant code; retained for result management and not used by the model.
sex	Participant sex; retained for result management and not used by the model.
age	Numeric age in years; every participant must be aged 65 years or older.
EQ5D	Numeric EQ-5D health utility index.
pa_aerobic	Meets the specified aerobic activity recommendation = 0; does not meet it = 1.
is_allownc	Receives livelihood assistance or subsidies = 1; does not receive them = 0.
Chew_Diff	Difficulty chewing = 1; no difficulty chewing = 0.
Prot_Deficiency	Protein intake is insufficient = 1; sufficient = 0. Insufficient means below 60 g/day for men and below 50 g/day for women.
N_EN	Numeric daily energy intake in kcal/day.
HE_wc	Numeric waist circumference in cm.
obe_4class	Normal weight = 0; underweight = 1; overweight = 2; obesity = 3.

Weight-status definitions

- Normal weight: 18.5 to less than 23 kg/m².
- Underweight: less than 18.5 kg/m².
- Overweight: 23 to less than 25 kg/m².
- Obesity: 25 kg/m² or greater.

Important: numeric codes must match those used in the model-development data. Do not replace category codes with free-text labels.

4. Upload and automated input checks

Open the Screening tab and select a supported batch file. Processing begins after the file is selected. The application checks the following conditions before prediction:

- all required columns are present;
- model inputs are numeric;
- age is present and at least 65 years;
- binary fields contain only 0 or 1;
- obe_4class contains only 0, 1, 2, or 3;
- participant identifiers are not missing or duplicated; and
- non-age model inputs are assessed for missing values.

Invalid data produce an error and no prediction is generated. Missing non-age model inputs produce a warning and are imputed by the locked preprocessing pipeline. Investigate missingness rather than treating imputation as a substitute for data-quality review.

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Intended Users

This tool is intended for trained community nurses, primary care physicians, public health practitioners, and qualified research staff conducting sarcopenia risk screening among adults aged 65 years and older.

Recommended Workflow

1. Collect and verify the required screening variables.
2. Remove unnecessary direct identifiers and upload a correctly formatted CSV or Excel file.
3. Review participants classified as screen positive.
4. Confirm findings using established assessments of muscle strength, muscle mass, and physical performance.
5. Refer participants for clinical assessment according to local protocols when appropriate.

Training Before Use

Users should understand the variable definitions and coding rules, the meaning of a screen-positive result, the limits of model-generated scores, and their local confirmatory assessment and referral procedures. The tool should only be introduced within an organization after an appropriate clinical or research lead has reviewed these requirements with users.

Figure 2. The How to Use tab provides the workflow, training requirements, input definitions, and a downloadable CSV template.

5. Review and download results

The results page reports the number screened, screen-positive and screen-negative counts, positivity rate, a summary chart, and a participant-level table. Download Screening Results (CSV) exports the complete result table.

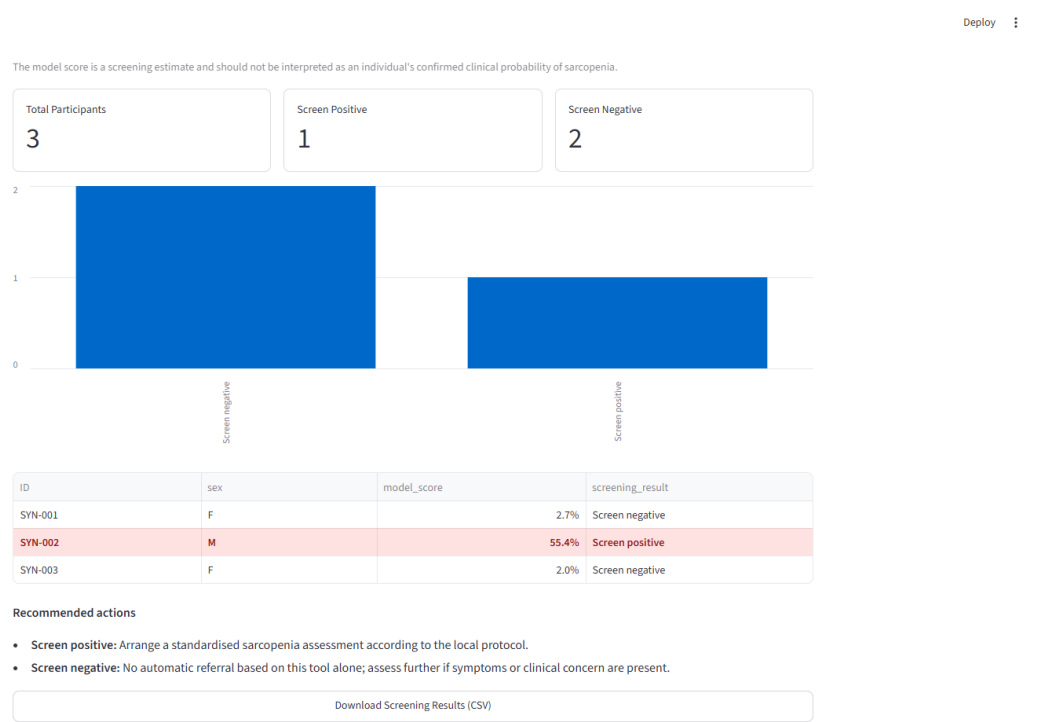


Figure 3. Batch results generated from synthetic demonstration records.

Output field	Meaning
model_score	Model-generated screening score. It is not a confirmed individual clinical probability.
screening_result	Screen positive or Screen negative, determined using the locked threshold.
recommended_action	Standardised next-step guidance associated with the screening classification.

6. Interpret screening classifications

Screen positive: Arrange a standardised sarcopenia assessment according to the local protocol.

Screen negative: No automatic referral based on this tool alone; assess further if symptoms or clinical concern are present.

Interpretation principles

- A screen-positive result is not a diagnosis.
- A screen-negative result does not prove that sarcopenia is absent.
- The model score is a screening estimate, not a confirmed individual clinical probability.
- Symptoms, functional decline, examination findings, and professional judgement remain relevant.
- Confirmatory assessment should follow an established local protocol.

Version separation

The locked model is version 1.0.0 and the web application is version 1.1.0. Interface, documentation, or software-validation changes do not imply that the model was retrained.

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Screening How to Use **Model Info** Governance

Model Information

Item	Details
Algorithm	LogisticRegression
Model version	1.0.0
Web application version	1.1.0
Current status	Research demonstration and preliminary screening aid
Web framework	Streamlit
Hosting platform	Render
Predictor set	Model_2
Number of predictors	9
Development sample	2022 dataset, n=889
Temporal validation sample	2024 dataset, n=1156
Screening threshold	0.082884 (8.2884%)
Threshold selection	Locked Youden threshold from 2022 OOF predictions in the latest notebook

Figure 4. Model and web-application version information and technical implementation details.

7. Temporal validation performance

The Model Info tab reports temporal validation using 2024 KNHANES data (n = 1,156). These estimates describe performance in the reported validation sample and do not guarantee equivalent performance in another population or setting.

Metric	Estimate	95% CI
ROC-AUC	0.837	0.792-0.877
PR-AUC	0.324	0.251-0.425
Accuracy	0.767	0.742-0.792
Balanced accuracy	0.776	0.729-0.822
Sensitivity	0.786	0.702-0.869
Specificity	0.766	0.740-0.791
PPV	0.208	0.183-0.235
NPV	0.979	0.970-0.987
F1 score	0.329	0.292-0.369
Brier score	0.058	0.054-0.061

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Threshold selection	Locked Youden threshold from 2022 OOF predictions in the latest notebook
scikit-learn version	1.6.1
Last updated	August 2026

Temporal Validation Performance

Performance of the locked final model in the 2024 KNHANES temporal validation dataset.

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Figure 5. Temporal-validation estimates and 95% confidence intervals displayed in the application.

8. Governance and privacy

The application code does not intentionally write uploaded participant files to persistent application storage. Files are processed during the active application session. Users must nevertheless assess the complete operational environment, including browser, network, hosting provider, access controls, downloads, and local storage.

Required safeguards

- Use non-identifying participant codes.
- Remove unnecessary direct identifiers.
- Restrict access to input and output files.
- Follow applicable institutional and data-protection requirements.
- Retain or delete downloaded results according to an approved data-management plan.

Prohibited uses: diagnosis, autonomous treatment decisions, replacement of professional judgement, or screening outside the intended population.

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Screening How to Use Model Info **Governance**

Intended and Prohibited Uses

The intended use is preliminary sarcopenia risk screening and research demonstration in community or primary care settings for adults aged 65 years and older. The tool must not be used to diagnose sarcopenia, make autonomous treatment decisions, replace professional judgment, or screen people outside the intended population.

Data Privacy

The application code does not intentionally write uploaded participant files to persistent application storage. Files are processed during the active application session. Users should remove unnecessary direct identifiers and comply with institutional policies, applicable data-protection requirements, and the hosting provider's terms before uploading any participant data.

Model Updating and Version Control

The model is not automatically retrained or updated. Any future revision requires documented validation, a new version number, updated performance and limitation statements, and review before deployment. The model version and update date shown on this page identify the currently deployed release.

Calibration and Performance Monitoring

Figure 6. Intended and prohibited uses, privacy, monitoring, version control, and clinical responsibility.

9. Troubleshooting

Message or problem	Recommended response
Unsupported file type	Save the file as CSV or XLSX and upload it again.
Missing required columns	Restore the exact required column names and check spelling and capitalisation.
Non-numeric values	Replace text in model-input fields with valid numeric values or codes.
Participant younger than 65	Remove the record; the model is restricted to adults aged 65 years and older.
Invalid binary code	Use only 0 or 1 according to the definition for that variable.
Invalid weight-status code	Use only 0, 1, 2, or 3 according to the defined BMI categories.
Missing model inputs warning	Verify the source data. If values remain missing, document that locked-pipeline imputation was used.
Missing or duplicate ID warning	Correct identifiers so results can be matched to the intended participant records.
Model files cannot be loaded locally	Install the pinned dependencies and confirm that all locked model artifacts are present.

Reproducibility and validation evidence

The public repository includes synthetic input and expected output, automated tests, pinned Python and package versions, locked model artifacts, and VALIDATION.md. Run the automated checks from the repository root using:

```
python -m unittest discover -s tests -v
```

Screenshot provenance

All screenshots in this manual were generated using the synthetic records in examples/sample_input.csv. They contain no participant or clinical-study data.

Maintenance

Review and update this guide whenever the web application version, input format, output format, or operational workflow changes.